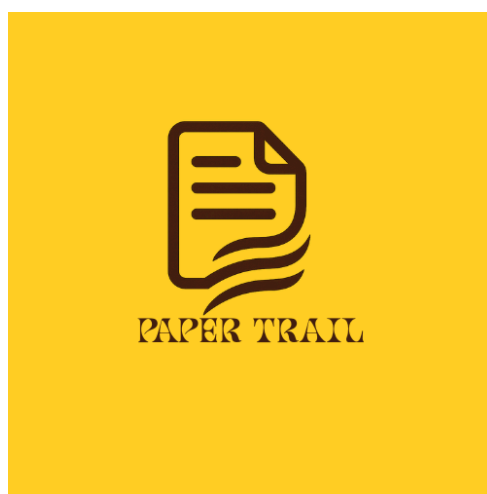




PaperTrail

User Guide

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V1.0

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1. Introduction

PaperTrail is an assistance software aimed to help users keep track of notes, important information, and constructive or general thoughts while using the online software known as Google Docs. This software aims to cut down on time spent searching for notes scattered across documents, letting users get back to developing documents instead of searching. The software runs in the background, allowing the user to bring up a sidebar to find tags generated by said user. This provides an intuitive and lightweight experience for all users, versus the confusing and unintuitive experience outlined by similar software like Asana and Notion. Customers will be able to easily incorporate the software into their Google Documents by simply going through Google's Extension Store. This User Guide will answer any questions from users about installing and running the software.

1.1 Intended Readership

- Researcher Specialists that already are experienced with using Google Docs should skip to the instruction section. Those that are not experienced should start with the overview section first.
- “Casual” Users would benefit from starting at the overview section to gauge their experience with Google Docs, and from there, determine whether they should continue to the instructions section.
- System Administrators, those concerned with system installation, should proceed to the instructions section. Those concerned with security should follow the reference section to gain additional information on how the software operates. Those concerned with error possibilities should head to the Error Messages and Recovery Procedures section.
- User Support Staff will spend most of their time going through the instructions section, as well as Error Messages and Recovery Procedures section to effectively assist users.

(non-native English speakers should request assistance with parsing this document)

1.2 Applicability Statement

The software released at the time of this User Guide creation will be version 1.0, relating to the User Guide version 1.0. This means that errors found after the release of the software won't be covered in this document.

1.3 Purpose

This User Guide is meant to be the absolute truth for information about the software. Answering the questions proposed by all users regardless of experience. This document is the baseline for all User Support Staff for assisting “Casual” Users on a corporate level. System Administrators will be able to follow instructions procedures for installation and upkeep of the software.

1.4 How to Use this Document

- The introduction section is intended to provide readers a quick overview of the document; guiding them to a section they need to critically read in a timely fashion.
- The overview section intends to provide a more detailed oriented section of how the system works over the introduction.
- The instructions section aims to instruct readers on the operations procedures, warnings, safety protocols, session upkeep and termination, and additional instructions for non “end” users concerned with the software itself.
- The reference section outlines the systems capabilities, specifically the functional description, cautions, and warnings. While also providing examples and possible error messages with potential causes.
- The error messages and recovery procedures sections elevates the reader on what sort of errors are possible from the software. Giving guidance on how to recover the software from said error with minimal damage.

1.5 Related Documents

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail System Requirements Specification*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail System Design Document*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail User Interface Design Document*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail Critical Design Review Document*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2026) *PaperTrail Code Inspection Report*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2026) *PaperTrail Administrator Manual*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2026) *PaperTrail User Guide*

GitHub Repository: <https://github.com/AnthonyVeilleux/PaperTrail>

1.6 Conventions

When a user is generating a tag, a bookmark flag will be automatically generated to the left of the tag, letting the user know the software properly parsed their tag. The sidebar allocates 20% of the user's screen space, allowing the user to still read the properties of the sidebar, while also being able to access the document. Important buttons are highlighted with background elements to convey their importance to the user. The search bar will have text present if the user hasn't inputted anything to convey that the search bar has interactivity.

1.7 Problem Reporting Instructions

Additional errors discovered should be addressed to the development team in terms of tickets via the GitHub repository, using the built in issues system. There the development team can address pressing issues with the software as time elapses.

2. Overview

PaperTrail is an assistance software that runs in the background when the user is using the service provided by Google known as Google Docs. The system automatically stores tags generated by the user without explicit command. Users can elect to open a sidebar, which presents them with a view of all tags found on the document. Users can click on a tag that will bring them to the location of the tag on the document. If multiple instances of the tag are present, then a dropdown will be displayed under the tag with a preview of text found under each tag, that can then be clicked to bring the user to the specific tag location. Tags can be nested in the form of objects (Ex. Tag.date) that will prompt the user more options for tag filtering and creation. A search bar is present that will query the document for tag names inputted by the user for quick navigation. There is another tab called “Global” which allows the user to find separate documents with tags specified in the search bar. If tags are found in the database, the featured document will be displayed, which the user can click on to open said document in a new tab. Finally, there is another tab called “Recent” which will display a timeline of created tags over the document's lifespan. Allowing the user to quickly pick up where they were on the document from their last tag created or accessed. In the dropdown menu for PaperTrail, there is an option to index a document into a database hosted in the cloud of the user's Google Sheets. There, users can elect which documents they want the global search tab to access for indexing purposes. The most the user will have to do is refresh the database before they put in a search query to guarantee they will find the tag they are searching for.

3. Instructions

This instruction section will outline each function the software is capable of performing. Instructing users on what the intended purpose of each operation, and possible defects/limitations of the software.

#1 Opening PaperTrail

To use PaperTrail, you must first open the extension within your Google Doc.

1. Open your desired document in Google Docs
2. In the top menu, click: Extensions - PaperTrail - Open Tag Sidebar
3. The PaperTrail sidebar will appear on the right side of your screen.

If the sidebar does not appear, make sure that you have properly installed the sidebar and have granted authorization

#2 Creating Tags

PaperTrail uses hashtags (#) to categorize and organize your notes within a given document.

A. Basic Tag Format

- a. Type a tag anywhere in your document.
- b. Ex. #Note

B. Nested Tags

- a. Tags can follow a hierarchical structure using a dot notation to indicate parent and child.
- b. #Note.Today

C. Dated Tags

- a. Users can attach a date to a tag using the following formats
- b. #Note [MM/DD] or #Note [M/D] depending on the month/day desired.

All tag formats are interchangeable and can be used together within the same document.

#3 Viewing and Managing Tags

Once tags are created, they will appear in the PaperTrail sidebar.

- Tags are automatically refreshed every 2 seconds
- If tags do not appear or seem outdated
 - Click “Refresh Tags” from the PaperTrail menu in the toolbar.

Sidebar

- View all created tags within your document
- Use the search bar to quickly find specific tags
- Click on the desired tag to navigate to the location of said tag.

#4 Understanding Sidebar Tabs

The PaperTrail sidebar is organized into tabs to help you filter and manage tags more efficiently

Local Tab:

- Primary View for most users
- Displays tags found in current document
- Use this tab to quickly browse and interact with tags you have created

Global Tab:

- View for searching for documents containing specific tags
- You can enter multiple tags at once. Results will show documents using AND/OR logic, as specified by the search query.
- The Global tab searches a pre-built index, not the documents in real time. If a document was recently edited, be sure to sync the indexed documents using the refresh button in the search bar or in the extension dropdown
- The Global tab only searches documents that have been added to the index scope. To add documents to scope see section #6.

Recent:

- Displays tags that include date information
- Automatically includes tags using the [MM/DD] or [M/D] format
- Useful for tracking time based notes, logs, or recent activity on the document if working with collaborators
- Filter using the 1 Day, 3 Day, 1 Week, or 1 Month depending on intent. Offers custom bounds for search.

#5 Using Tag Autocomplete

PaperTrail provides autocomplete functionality in the sidebar in order to improve efficiency.

- Begin typing in the sidebar searchfield
- Suggested tags will appear based on the existing tags within the document
- Select a suggestion to quickly locate the tag

Note: Due to Google limitations, autocomplete is only available in the sidebar and not directly within the document editor

#6 Exporting Tagged Notes

PaperTrail allows the user to export their tagged content for sharing or reusing.

Export Options:

1. Export to a new Google Doc
2. Export as a PDF

How to Export:

1. Open the PaperTrail menu
2. Select **Export Tagged Notes**
3. Choose your desired Tag to export
4. Choose your desired output format
5. Click Export

#7 Managing Document Scope for Global View

1. Open the PaperTrail menu
2. Select Index Scope -> Manage Scope

Including a folder

To quickly add the folder containing your current document, click *Use Current Doc Folder* instead, or paste a Google Drive folder URL or ID into the *Include A Folder* field and click *Include Folder*. Every Google Doc inside that folder will become searchable from the Global tab.

Including a single document

To add the document you currently have open, click *Use Current Doc*, or paste a Google Doc URL or ID into the *Include A Single Document* field and click *Include Doc*.

Removing a folder or document

Each included item appears in the list at the bottom of the dialog with a *Remove* button. Click it to remove that item from the scope.

Syncing the index

Adding or removing items does not update the search index immediately. Click *Sync Index* (or *PaperTrail > Sync Indexed Docs*) after making changes. The dialog will show how many documents were updated, skipped, or failed.

4. Reference Section

This section provides comprehensive reference information for every PaperTrail operation. Each subsection covers: functional description, cautions and warnings, formal parameter description, examples, possible error messages with causes, and cross-references to related operations.

4.1 Opening PaperTrail (showSidebar)

4.1.a Functional Description

The Open Tag Sidebar button opens the PaperTrail sidebar inside the active Google Doc. Internally it calls `showSidebar()`, which renders `Index.html` as a templated sidebar at a fixed width approximately 20% of the document environment to the right side of the browser window. The sidebar displays the Local tab by default and begins polling for tag data every 2 seconds automatically.

4.1.b Cautions and Warnings

PaperTrail must be installed from the Google Workspace Marketplace before this option is available in the Extensions menu.

On first use, Google will prompt for authorization of the OAuth scopes listed in `appsscript.json` (documents, drive, spreadsheets, script UI, user email). The sidebar will not open until authorization is granted.

A keyboard shortcut tip dialog appears automatically on every document open (`onOpen`). It can be dismissed normally and does not affect sidebar functionality.

4.1.c Formal Description

Menu path	Extensions > PaperTrail > Open Tag Sidebar
Required	An open Google Doc; PaperTrail installed and authorized
Optional parameters	None
Default tab shown	Local (filter: 'projects')
Auto-refresh interval	2 seconds (<code>startAutoRefresh()</code> in <code>Scripts.html</code>)
Result	Sidebar opens on the right; tag loading spinner displays, then Local tab renders

4.1.d Examples

To open the sidebar on a new document:

1. Click Extensions in the Google Docs menu bar.
2. Hover over PaperTrail.
3. Click Open Tag Sidebar.
4. The sidebar appears on the right. A loading spinner shows briefly while `getAllTags()` fetches tag data from the document.

4.1.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
Sidebar does not appear	Extension is not installed or authorization was denied. Reinstall from the Google Workspace Marketplace and re-authorize when prompted.
Extensions menu has no PaperTrail item	The add-on is not installed for this Google account or browser profile. Install from the Marketplace.
Sidebar opens but shows only spinner	<code>getAllTags()</code> may have thrown an error. The most common cause is an expired authorization token; re-authorize via Extensions > Manage Add-ons.

4.1.f Cross-References

See Section 4.2 for creating tags once the sidebar is open. See Section 4.3 for sidebar navigation. See Section 4.4 for tab descriptions.

4.2 Creating Tags

4.2.a Functional Description

Tags are hashtag-prefixed labels typed directly into the Google Doc body. PaperTrail detects them using the regular expression (regex pattern). No save command is needed; tags are picked up automatically during the next sidebar refresh cycle. Three formats are supported and may be mixed freely within the same document.

4.2.b Cautions and Warnings

- Tag names may only contain letters (a-z, A-Z), digits (0-9), underscores (_), hyphens (-), and dots (.). A space after # breaks detection (e.g. # Note is not a valid tag).
- Tag names are case-sensitive: #Note and #note are stored as two distinct tags.
- The dot (.) is reserved for nesting. A tag like #v1.0 is parsed as parent 'v1' with child 'v1.0'.
- Deleting a tag from the document body removes it from the sidebar on the next auto-refresh or manual Refresh Tags.

4.2.c Formal Description

Basic Tag

Syntax	#TagName
Required	# character immediately followed by one or more valid characters (no spaces)
Optional	None
Examples	#Note #Todo #ActionItem #v1_final
Result	Tag appears in the sidebar Local tab with a colored dot and occurrence count badge. A bookmark is created at each occurrence location.

Nested Tag (dot notation)

Syntax	#Parent.Child or #Parent.Child.GrandChild
Depth supported	Unlimited levels (each dot adds one level)
Required	Valid parent name, a dot separator, and a valid child name
Storage	Parent-child relationship stored in hierarchyMap and persisted via establishNestedTagRelationships() in TagService.js
Examples	#Note.Today #Project.Alpha #Research.Paper.Section2
Result	Parent appears in the sidebar as a collapsible group node; child tags appear indented beneath it.

4.2.d Examples

Basic: Type #Todo in any paragraph. After the next 2-second refresh a bookmark flag appears to the left of the text in the document, and #Todo appears in the sidebar with a count badge.

Nested: Type #Project.Alpha. After the next refresh, Project appears as a parent group in the sidebar with Alpha listed as a child item. Clicking Project expands or collapses the child list.

4.2.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
No Tags Found (alert dialog)	Informational only - not an error. The document contains no valid hashtags. Add a tag using one of the formats above and click Refresh Tags.
Tag does not appear in sidebar	The 2-second auto-refresh has not yet run, or the tag format is invalid (e.g., space after #). Verify the format, then click Extensions > PaperTrail > Refresh Tags.
Bookmark flag missing in document	ensureTagBookmarks() may not have run yet. Click Refresh Tags to force a rebuild of all bookmarks.
#v1.0 appears as two separate tags	The dot is always treated as a nesting separator. Rename to #v1_0 if nesting is not intended.

4.2.f Cross-References

See Section 4.3 for navigating to tags via the sidebar. See Section 4.4 for the Recent tab and date filtering. See Section 4.6 for exporting tagged content.

4.3 Viewing and Managing Tags (Sidebar - Local Tab)

4.3.a Functional Description

The sidebar Local tab displays all tags found in the current document, grouped into a Default Project. Each tag shows a colored dot, the tag name, and an occurrence count badge. Clicking a single-occurrence tag calls `jumpToTagBookmark(tagName, 0)` to move the document cursor to that location. Tags with multiple occurrences display a circular toggle button; clicking it expands a dropdown of snippet previews fetched via `getTagOccurrences()`, each of which can be clicked to navigate to that specific occurrence. The sidebar auto-refreshes every 2 seconds by polling `getAllTags()` and comparing the `lastUpdated` timestamp.

4.3.b Cautions and Warnings

- Auto-refresh polls every 2 seconds. Changes made to the document body may take up to 2 seconds to appear in the sidebar.
- The sidebar occupies 360 px of the screen. On smaller screens, document readability may be reduced.
- The search bar in the Local tab filters by tag name only, not by document body text.
- Occurrence snippets show up to 80 characters of text following the tag. Lines consisting only of hashtags or separators are skipped automatically.

4.3.c Formal Description

Auto-refresh interval	2 seconds
Manual refresh	Extensions > PaperTrail > Refresh Tags
Tag navigation - single occurrence	Click the tag row > jumpToTagBookmark in the document
Tag navigation - multiple occurrences	Click the circular count button to expand snippet list; click a snippet > jumpToTagBookmark(tagName, index)
Snippet length	Up to 80 characters from the line following the tag, or the next meaningful sibling paragraph
Search / filter	Type in the search bar to filter the Local tab list by tag name (case-insensitive substring match)
Tag color	Assigned randomly from 6 colors on first detection; stored in document properties under key tag_<TagName>
Nested tag expand/collapse	Click the chevron icon next to a parent tag to show or hide its children

4.3.d Examples

A 20-page document has #Todo appeared 5 times. In the sidebar, clicking the '5' badge next to #Todo expands 5 snippet previews showing surrounding text for each occurrence. Clicking the third preview calls jumpToTagBookmark('Todo', 2), moving the cursor to the third occurrence.

4.3.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
Tags seem outdated	Auto-refresh delay or network latency. Click Refresh Tags manually from the PaperTrail menu.
Failed to refresh tags: <detail>	An unexpected error in getAllTags() or upsertDocumentTagsToIndex_(). Note the detailed text, close and reopen the document, and try Refresh Tags again. If a quota or rate limit is cited, wait a few minutes.
Occurrence snippet shows 'No preview text available'	The tag is followed only by blank lines or other hashtags. Add text content after the tag heading to generate a preview.

4.3.f Cross-References

See Section 4.4 for the Global and Recent tabs. See Section 4.5 for autocomplete in the search bar.

4.4 Sidebar Tabs: Local, Global, and Recent

4.4.a Functional Description

The sidebar header contains a two-button tab bar. The Local tab (filter: 'projects') shows tags in the current document. The Global tab (filter: 'global') searches the cross-document spreadsheet index. The Recent tab is a sub-view of the Local tab showing only dated tags filtered by time range.

4.4.b Cautions and Warnings

- The Global tab queries a pre-built Google Sheets index (PaperTrail_Index spreadsheet, TagIndex sheet). It does not scan documents in real time. Always sync the index after editing indexed documents.
- Global search uses AND logic by default: a document must contain ALL queried tags to appear in results.
- The Global tab only returns documents that have been added to Index Scope AND synced. Documents outside scope are invisible to Global search.
- The Recent tab only displays tags that include the [MM/DD] or [M/D] date format. Tags without dates never appear on the Recent tab.

4.4.c Formal Description

Local Tab

Filter value	'projects'
Data source	getAllTags() - scans the active document body in real time
Default tab	Yes - shown immediately on sidebar open
Search behavior	Case-insensitive substring match on tag names in the current document
Autocomplete	Suggests matching tags from the current document as the user types

Global Tab

Filter value	'global'
Data source	searchIndexedDocsByTags() - queries the PaperTrail_Index Google Spreadsheet
Search logic	AND (all entered tags must be present in a document for it to appear)
Multi-tag input	Enter multiple tags separated by spaces; autocomplete works per-token
Sync required	Click the refresh icon in the Global search bar, or Extensions > PaperTrail > Sync Indexed Docs
Result display	Each matching document shown as a card with title, matched tag chips, tag count, and an Open link
Autocomplete source	getIndexedTagSuggestions() - reads from the index spreadsheet

Recent Tab

Access	Click the Recent section inside the Local tab view
Data source	Subset of Local tab tags that include [MM/DD] or [M/D] date notation
Filter buttons	1 Day, 3 Day, 1 Week, 1 Month
Use case	Time-based notes, daily logs, tracking recent activity on collaborative documents

4.4.d Examples

Global tab example: A user enters '#Action #Urgent' in the Global search bar and clicks the search button. PaperTrail calls `searchIndexedDocsByTags(['Action','Urgent'], 'AND')`. Three documents appear, each confirmed to contain both tags. The user clicks Open on one result to open it in a new browser tab.

Recent tab example: A document contains #Meeting [04/10], #Meeting [04/12], and #Review [04/05]. On the Recent tab with the 1 Week filter, only the April 10 and April 12 entries appear (assuming today is April 13).

4.4.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
No results in Global tab	No documents in scope contain all queried tags, or the index is stale. Sync the index (Extensions > PaperTrail > Sync Indexed Docs) and retry.
Index Sync Completed With Errors	One or more documents in scope could not be read or indexed, usually due to insufficient permissions. Check that the Google account has at least Viewer access to all documents in scope. Re-run Sync Indexed Docs after resolving permissions.
Index Sync Failed / <error detail>	A fatal error stopped the entire sync. Confirm at least one folder or document has been added via Index Scope > Manage Scope. Recheck folder IDs for typos, then re-run Sync Indexed Docs.
Recent tab shows no tags	No tags in the document use the [MM/DD] or [M/D] date format. Add a dated tag (e.g., #Note [04/13]) and refresh.

4.4.f Cross-References

See Section 4.6 for managing Index Scope. See Section 4.2 for creating dated tags used by the Recent tab. See Section 4.5 for autocomplete behavior in the Global tab.

4.5 Tag Autocomplete

4.5.a Functional Description

As the user types in the sidebar search field, PaperTrail displays a dropdown of matching tag suggestions. On the Local tab, suggestions are drawn from the current document's tag list (`getAllTagsFromData()`). On the Global tab, suggestions are drawn from the cross-document index (`getIndexedTagSuggestions()`), loaded once on first use and cached in memory.

Autocomplete is available in the sidebar only; it is not available directly in the Google Doc editor body due to a Google platform limitation.

4.5.b Cautions and Warnings

- Autocomplete is only available inside the sidebar search field, not in the document body. This is a limitation of the Google Apps Script HTML service.
- On the Global tab, autocomplete suggestions load from the index spreadsheet on first use. If the index has not been synced recently, suggestions may not reflect the latest tags.
- Up to 10 suggestions are shown at a time.

4.5.c Formal Description

Trigger	Any character typed in the sidebar search field
Local tab source	getAllTagsFromData() - in-memory list from the current document
Global tab source	getIndexedTagSuggestions(prefix, 10) - reads from the PaperTrail_Index spreadsheet
Match type	Case-insensitive substring match (query appears anywhere in tag name)
Max suggestions shown	10
Keyboard navigation	Arrow Up / Arrow Down to move selection; Enter to accept; Escape to dismiss
Space key (Global tab)	Accepts the current suggestion and appends a space, allowing the user to type additional tag tokens
Click	Clicking any suggestion item navigates to that tag (Local) or inserts it into the Global search field
Dismiss	Click anywhere outside the dropdown, or press Escape

4.5.d Examples

Local tab: A document contains #Note, #Note.Today, and #NoteImportant. The user types 'no'. All three appear as suggestions with occurrence counts. The user presses Arrow Down twice to highlight #Note.Today and press Enter - the sidebar navigates to that tag.

Global tab: The user types 'act'. The dropdown shows #Action (14), #ActionItem (3) drawn from the index. The user presses Space to accept #Action and continue typing 'urg' to add a second token before pressing Enter to run the AND search.

4.5.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
No suggestions appear (Local tab)	No tags in the current document match the typed characters. Verify tags exist using the full Local tab list.
No suggestions appear (Global tab)	The index spreadsheet is empty or has not been synced. Run Extensions > PaperTrail > Sync Indexed Docs and retry.

4.5.f Cross-References

See Section 4.3 for Local tab behavior. See Section 4.4 for Global tab search. See Section 4.6 for syncing the index that powers Global autocomplete.

4.6 Exporting Tagged Notes (exportTaggedNotes)

4.6.a Functional Description

The Export Tagged Notes feature extracts all paragraphs and list items that fall under a selected tag heading and saves them to a new Google Doc or a PDF file in the user's Drive. Internally, collectParagraphBlocksWithTag() scans the document body sequentially; a block starts at any paragraph beginning with #TagName and ends when the next #OtherTag paragraph is encountered. For PDF export, exportBlocksAsPdf_() creates a temporary Google Doc, styles it, converts it to PDF, saves the PDF to Drive, and then trashes the temporary Doc.

4.6.b Cautions and Warnings

- Only content physically beneath a tag heading paragraph is exported. Content that appears before any tag heading in the document is never included.
- Export creates a new file in the user's Google Drive; it does not modify the original document.
- PDF export creates a temporary Google Doc in Drive, pauses 1.5 seconds for rendering, converts to PDF, then marks the temp doc as trashed. If the script times out, the temp doc may remain in Drive.
- Duplicate content blocks (identical text) are deduplicated before export.
- If no notes exist under the selected tag, the operation throws: 'No notes found for #<TagName>'.

4.6.c Formal Description

Menu path	Extensions > PaperTrail > Export Tagged Notes
Backend entry point	exportTaggedNotes(tagName, format) - ExportService.js
Required parameters	tagName (string, without #); format ('DOC' or 'PDF')
Output: Google Doc	New document created via DocumentApp.create(); returns {url, id, type:'DOC'}
Output: PDF	New PDF file created via DriveApp.createFile(blob); returns {url, id, type:'PDF'}
Tag list source	getTagNamesForExport() calls extractHashtagsFromDocument() and returns sorted tag name array
Block detection regex	Regex pattern applied to trimmed paragraph text
Deduplication	Identical exported blocks are removed; order is preserved

Export Procedure:

1. Click Extensions > PaperTrail > Export Tagged Notes.
2. In the dialog, select the desired tag from the Tag dropdown.
3. Select the output format: Google Doc or PDF.
4. Click Export Notes.
5. When complete, a link to the exported file appears in the dialog. Click Open exported file to view it.

4.6.d Examples

A researcher has tagged meeting notes with #Meeting throughout a 30-page document. They open Export Tagged Notes, select #Meeting, choose PDF, and click Export. After a few seconds, a link to a styled PDF appears. The PDF contains a header block (document name, tag, export timestamp) followed by numbered entries, each beginning with the #Meeting heading line and the content that followed it.

4.6.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
No tags found in this document	The document contains no valid hashtags. Create at least one tag before attempting export.
No notes found for #<TagName>	The selected tag exists in the document but has no content paragraphs beneath it. Add text after the tag heading and retry.
Export failed: <detail>	A Google Drive permission or quota issue prevented file creation. Check available Drive storage and that the account has write access to Drive, then retry.

4.6.f Cross-References

See Section 4.2 for creating properly structured tag headings. See Section 4.3 for reviewing tag occurrences before export.

4.7 Managing Document Scope for Global Search**4.7.a Functional Description**

Index Scope controls which Google Docs are searchable from the Global tab. Scope is stored in Script Properties. When Sync Index Now is triggered, the backend will enumerate all Google Docs in each configured folder, merges the explicit document list, skips files whose MD5 text signature is unchanged, and upserts tag rows into the PaperTrail_Index spreadsheet under the TagIndex sheet. The backend has a function called LockService.getScriptLock() that prevents concurrent index writes.

4.7.b Cautions and Warnings

- Adding or removing items from scope does NOT update the search index immediately. Always click Sync Index Now after any scope change.
- The Google account running PaperTrail must have at least Viewer access to all documents and folders in scope.
- Scope is stored in Script Properties, which are shared across all users of the same script deployment. Changes one user makes to scope affect all users.
- Folder scope is not recursive: only the top-level files in a folder are indexed. Subfolders must be added separately.
- Removing a folder or document from scope does not delete it from Drive. The removed items remain searchable until the next sync.

4.7.c Formal Description

Menu path	Extensions > PaperTrail > Index Scope > Manage Scope
Add a folder (URL or ID)	Paste a Google Drive folder URL or raw folder ID into Include A Folder > click Include Folder. Calls addIndexContentFolder(input), which normalizes via normalizeDriveFolderId_() and appends to PAPERTRAIL_CONTENT_FOLDER_IDS.
Add current doc's folder	Click Use Current Doc Folder. Calls addCurrentDocumentParentFolderToIndexScopeForUi(): reads the first parent folder of the active document.
Add a single document (URL or ID)	Paste a Google Doc URL or raw doc ID into Include A Single Document > click Include Doc. Calls addIndexExplicitDocument(input), which normalizes via normalizeGoogleDocId_() and appends to PAPERTRAIL_EXPLICIT_DOC_IDS.
Add current document	Click Use Current Doc. Calls addCurrentDocumentToIndexScopeForUi() - adds the active document's ID directly.
Remove a folder	Click Remove next to the folder entry. Call removeIndexContentFolderId(folderId).
Remove a document	Click Remove next to the document entry. Call removeIndexExplicitDocId(docId).
Sync the index	Click Sync Index Now in the dialog, or Extensions > PaperTrail > Sync Indexed Docs. Calls backfillContentFolderToIndex().
Sync result fields	Scanned (total docs processed), Updated (rows written), Skipped (unchanged signature), Failed (access errors), Duration (ms)

ID normalization	Both folder and document inputs accept full Google URLs or bare IDs. Bare IDs must be at least 20 alphanumeric/dash/underscore characters.
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4.7.d Examples

Scenario: A team wants all documents in a shared project folder indexed for Global search.

1. Open Extensions > PaperTrail > Index Scope > Manage Scope.
2. Paste the Drive folder URL into the Include A Folder field.
3. Click Include Folder. A status message confirms the folder was added.
4. Click Sync Index Now. The dialog shows: Scanned: 12, Updated: 12, Skipped: 0, Failed: 0.
5. Switch to the Global tab in the sidebar and search for #Action. Results now include documents from that folder.

4.7.e Possible Error Messages and Causes

Error / Symptom	Diagnosis & Recovery
Failed to add folder: <detail>	The folder ID or URL was invalid, or the account lacks access. Paste the full Drive URL or confirm the ID is correct (the string after /folders/ in the URL). Verify the account has at least Viewer access.
Failed to add parent folder: <detail>	The active document has no parent folder, or the account cannot access it.
Failed to add current document / Failed to add document: <detail>	The document ID or URL was invalid, or the account lacks access. Use the full Google Doc URL (the string after /d/ and before /edit).
Index Sync Completed With Errors (Failed: N)	N documents in scope could not be read. Check permissions on flagged files and re-run sync after granting access.
Index Sync Failed / <error detail>	A fatal error stopped the entire sync. Verify Manage Scope shows at least one item, recheck IDs, and retry.
Access check failed (shown on scope list item)	getIndexScopeState() could not access the folder or document. The account no longer has access to that item. Remove it from scope or re-grant access.

4.7.f Cross-References

See Section 4.4 for using the Global tab after scope is configured. See Section 5 for full error recovery procedures across all operations.

5. Error Messages and Recovery Procedures

This Error Messages and Recovery Procedures section will assist users on how to optionally recover the system in the event of failures.

5.1 Google Docs Menu Alerts

Error: ***"No Tags Found"***

Diagnosis: Informational — the document contains no hashtags. Not a failure.

Recovery: Enter tags into the document, e.g. #Tagname, #Todo

Error: ***"Failed to refresh tags: <detail>"***

Diagnosis: An unexpected failure occurred while scanning the document or writing to the cross-document index. The detail text identifies the specific cause (quota exceeded, permissions, etc.).

Recovery: Note the detailed text. Close and reopen the document, then try Refresh Tags again. If the error references a quota or rate limit, wait a few minutes before retrying.

Error: ***"Index Sync Completed With Errors"***

Shows counts: Scanned / Updated / Skipped / Failed / Duration

Diagnosis: The sync finished but one or more documents in the configured scope could not be read or indexed, usually due to insufficient permissions on specific files.

Recovery: Check that the Google account running the add-on has at least Viewer access to all documents and folders in scope. Documents that failed are skipped; successfully indexed documents are unaffected. Re-run PaperTrail > Sync Indexed Docs after resolving permissions.

Error: ***"Index Sync Failed"*** / <error detail>

Diagnosis: A fatal error stopped the entire sync. Common causes: no scope configured, an invalid folder ID, or a Drive API failure.

Recovery: Confirm at least one folder or document has been added via Index Scope > Manage Scope. Recheck folder IDs for typos, then re-run Sync Indexed Docs.

Error: "***Failed To Add Folder***" / "***Failed To Add Parent Folder***"
"Failed To Add Current Document" / "***Failed To Add Document***"

Diagnosis: The provided ID or URL was invalid, or the account does not have access to the resource.

Recovery: Paste the full Google Drive URL or the raw ID (the string after /folders/ or /d/ in the URL). Confirm the account has at least Viewer access to the folder or document.

5.2 Sidebar – Tag List

Error: "***Global Search Failed***" + <error message>

(Shown in place of search results)

Diagnosis: The index search call failed outright. Common causes: the PaperTrail_Index spreadsheet was deleted, permissions on the spreadsheet changed, or a Drive API error occurred.

Recovery: Click Try Again. If the error persists, run PaperTrail > Sync Indexed Docs. If the index spreadsheet was deleted, PaperTrail will create a new one — note that all previous cross-document index history will be lost.

Error: "***Search timed out. Try syncing the index, then retry.***"

Diagnosis: The cross-document index search did not complete within the allotted time. Usually caused by a large index spreadsheet or a slow Drive API response.

Recovery: Run PaperTrail > Sync Indexed Docs to rebuild and compact the index, then retry the search.

Error: "***No Indexed Matches***" (no results returned for query)

Diagnosis: Informational — no documents in the index contain all of the searched tags. Not an error.

Recovery: Check the tag spelling. Run PaperTrail > Sync Indexed Docs to ensure recent documents are indexed, then try again.

5.3 Manage Scope Dialog

Error: "***Failed to load scope: <detail>***"

Diagnosis: The dialog could not read the configured folder and document list when it opened. Usually a temporary Apps Script connectivity issue or an expired authorization.

Recovery: Close and reopen the dialog. If the error persists, go to Extensions > Apps Script, re-authorize the add-on, and try again.

Error: "***Failed to add folder: <detail>***"

Diagnosis: The folder URL or ID could not be resolved. Common causes: value copied incorrectly, folder does not exist, or the account does not have access.

Recovery: Copy the folder URL directly from the browser address bar while the folder is open in Google Drive, or paste only the folder ID (the characters after /folders/ in the URL). Confirm the account has at least Viewer access.

Error: "***Failed to add document: <detail>***"

Diagnosis: The document URL or ID could not be resolved. Same causes as above.

Recovery: Copy the document URL directly from the browser, or paste only the document ID (the characters between /d/ and the next / in the URL). Confirm the account has at least Viewer access.

Error: "***Failed to add current document: <detail>***"

Diagnosis: PaperTrail could not identify or access the currently open document. Typically occurs when the add-on lost its session context.

Recovery: Close the dialog, reload the Google Doc page, reopen the sidebar, and try again via Index Scope > Manage Scope > Use Current Doc.

Error: "***Failed to add current document parent folder: <detail>***"

Diagnosis: The parent folder of the current document could not be accessed, usually because it is in a Shared Drive or a folder the account does not own.

Recovery: Open the folder in Google Drive directly and use Include A Folder with its URL instead.

Error: "***Failed to remove folder: <detail>***" "***Failed to remove document: <detail>***"

Diagnosis: The removal could not be written back to the configuration store. Usually a temporary Apps Script quota or write error.

Recovery: Wait a moment and click Remove again. If the error repeats, reopen the dialog — the item may have already been removed on

a prior attempt.

Error: **"Access check failed"** (shown on a listed item, not the status bar)

Diagnosis: The item is in the configuration but the account currently running the add-on cannot access it in Google Drive. Its contents will not be included when the index is synced.

Recovery: In Google Drive, confirm the account has at least Viewer access to the item. If access was intentionally revoked, click Remove to clean it from the scope list.

Error: **"Index sync failed: <detail>"**

Diagnosis: The sync could not complete at all. Common causes: no scope configured, a Drive API error, or a script timeout on a very large folder.

Recovery: Confirm at least one folder or document is listed before syncing. If the scope is configured correctly, wait a minute and click Sync Index Now again.

Error: **"Index sync complete. Updated: X, skipped: Y, failed: Z"** (with failed > 0)

Diagnosis: The sync finished but one or more documents could not be indexed. Documents that failed are skipped without affecting the rest of the index.

Recovery: Open Extensions > Apps Script > Executions to identify which document IDs failed. Typically this is a permissions issue on specific files within an indexed folder.

5.4 Export Dialog

Error: **"Select a tag first."**

Diagnosis: Informational — the Export button was clicked without choosing a tag. Not a system error.

Recovery: Select a tag from the list before clicking Export.

Error: **"Error loading tags: <detail>"**

Diagnosis: The tag list could not be fetched when the Export dialog opened. Usually the same underlying cause as "Error Loading Tags" in the sidebar (see Section 5.2).

Recovery: Close the dialog, reload the sidebar, and try reopening the Export dialog via PaperTrail > Export Tagged Notes.

Error: "***Export failed: <detail>***"

Diagnosis: The export process encountered an error writing the output document or reading the source content.

Recovery: Note the detailed text. Ensure there is sufficient Google Drive storage quota and that the account has permission to create new files in Drive. Retry the export.

Appendix A – Team Review Sign-off

All members of the PaperTrail Development Team have thoroughly reviewed this User Guide Document and agree that it accurately represents the project's goals, scope, and requirements. Each team member confirms that the content and format of this document meet the expectations of both the team and the customer. While minor editorial or stylistic differences may exist, there are no major points of disagreement among the team regarding the material presented herein.

Signatures

Name (Typed)	Signature	Date
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Appendix B – Document Contributions

The following table identifies each team member's contributions to the creation of this User Guide Document. Each member has contributed to writing, editing, and reviewing the document. The percentages represent an estimate of each member's contribution to the total effort.

Team Member	Contributions	Percentage of Work
Vasu Patel	Completed section 4 - reference section.	20%
Ben Yandell	Completed Section 3 - Instructions	20%
Robert Kulow	Completed section 1, implemented peer review, and handled signature procedures.	20%
Arius Ahmad	Worked on document formatting and section organization, as well as the table of contents.	20%
Anthony Veilleux	Completed Managing Document Scope for Global View section. Completed Section 5	20%